

THE NEEDLE TELEGRAPH

In 1837, Englishmen William Cooke and Charles Wheatstone patented the first practical telegraph. Activated by electric currents, its 5 needles could point to 20 letters of the alphabet to spell out received messages. By 1839, it was in use on the railways.

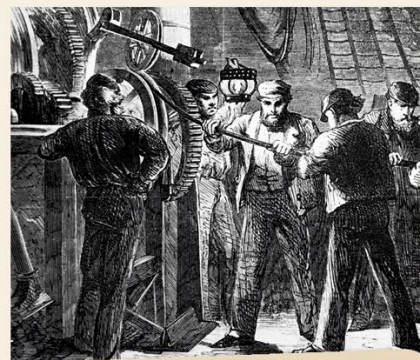


Cooke and Wheatstone's electric telegraph

Keys pressed in pairs to send letters

WOW!

The first words transmitted by Morse's telegraph were, "What hath God wrought", sent between Baltimore and Washington, D.C.



CROSSING BORDERS

The first transatlantic telegraph cable was laid under the sea between Ireland and Newfoundland, Canada in 1858, by the SS *Great Eastern* ship, above. This reduced communication time between Europe and North America from 10 days (a ship's crossing time) to just 17 minutes.

SENDING MESSAGES BY ELECTRICITY

Morse and Vail created a machine that worked by pushing the operator key down to complete the electric circuit of the battery. This sent the electric pulse along a wire to a receiver at the other end. Here, a small electromagnet powered by the electric pulse attracts a stylus that marks a paper tape with either short or long marks, representing dots and dashes.

